

Canonical Cofactor Provenance at a Fixed Shift: Minimal Faithful Relations, Exact Exposure Factorization, and the Single-Band Completion Barrier

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Abstract

The fixed-shift packet developed in TPC-47–TPC-62 eventually uses a finite retained relation $\mathcal{I}_X$ of source rows and cofactors. The earlier residual formulas, however, quantify over already supported (m, r) -pairs, while the later coefficient space rectangularizes the relation across orbit and opposite-row labels. No displayed theorem had identified that pair relation with the literal full incidence from which the physical terminal operator is built. Consequently, a favorable cofactor-exposure estimate cannot be inferred from the inherited interface without a circular support assumption.

We repair this logical provenance interface at one fixed nonzero shift. For an arbitrary finite literal incidence, we prove exact criteria for when pair projection and rectangular completion preserve the pre-terminal raw form, the terminal atomic map, or the grouped physical operator. The three criteria are different. We construct the unique minimal terminal-faithful pair relation when it exists, identify the spurious-cross obstruction when it does not, and show that the full incidence hypergraph is then the canonical replacement. Applied to the native very-high carrier, this gives a coefficient-independent maximal eligible relation, an effective-core identity, and the exact row factorization

$$\frac{W_R^T(m)}{W_H(m)} = \alpha_m \theta_m,$$

with an explicit zero-alignment convention. It separates terminal-support alignment from retained-relation completion and yields sharp uniform and distinguished exposure constants.

For the literal single-band provenance $\sqrt{N_+} < p \leq N_-/R_0$, intersection with the critical very-high face $p > y$ is confined to $R_0 \leq r < N_+/y$, up to one fixed-ratio transition window. At $y \asymp Q$, the balanced band, and in fact every band certified by the inherited freezing inequality, has $R_0 \gg J$ and is eventually disjoint from the critical face. Retuning to $R_0 \asymp J$ makes the old coefficient-blind freezing loss equal to $15077/105000 > 1/400$. Even after an independently justified retuning, every omitted full dyadic block below the top window makes the inherited Brun–Titchmarsh harmonic certificate yield zero fixed-power saving. A new multiscale construction or a weighted largest-prime-factor/terminal distribution theorem is therefore necessary for this support-only exposure branch. The result is an L1 provenance-completion criterion and a sharp closure of two support-only certificates; it is not a fixed-shift parity estimate, a prime-pair lower bound, or a twin-prime result.

Keywords: fixed shift; cofactor provenance; retained relation; terminal incidence; exposure factorization; support obstruction.

Convention. Every physical specialization uses one fixed nonzero shift h_0 , the literal coefficient-free structural support before the row coefficient is chosen, and raw counting-measure atomic norms. The symbol h_0 is not specialized to 2. A support identity is never promoted to a

prime-pair or parity statement.

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1 Introduction

The fixed-shift route preceding this paper has reached a precise but unresolved reassembly problem. TPC–59 decomposes the literal Walsh carrier into base, represented high, and missing high coordinates; TPC–60 identifies the exact shorted-Gram and kernel tests for restoring the missing high image; TPC–61 disintegrates its raw mass into realized row cofactor measures; and TPC–62 arranges all available closure certificates in the correct quantifier order [8–11]. The favorable mass–degree route would require, in matching scope,

$$\mathcal{D}_M(z) \leq X^{-\sigma_M + o(1)} \mathcal{D}_{\text{ref}}(z), \quad \sigma_M > \lambda_{\text{rep}} + \nu_M. \quad (1)$$

The first inequality is not a formal consequence of the definitions. It depends on which cofactors the inherited relation $\mathcal{I}_X$ actually retains on every active row.

That relation has an unresolved provenance. TPC–47 and TPC–48 begin with literal sums over already supported (p, r, m, j) -atoms and prove coefficientwise signed reassembly and fixed-row r -orthogonality [2, 3]. TPC–51 first names a set $\mathcal{I}_X$ containing the supported (m, r) -pairs; TPC–52 and TPC–53 then form a direct sum over that relation, and TPC–54 explicitly permits arbitrary finite cofactor multiplicity [4–7]. None of those displayed interfaces supplies a set-builder membership rule or a row-completion theorem. TPC–61 therefore correctly proves that the finite-relation hypothesis alone implies no positive exposure, but leaves open whether the literal upstream incidence repairs that defect.

This paper opens precisely that box. Its first point is conceptual but mathematically exact: a pair relation and a full four-label incidence are not interchangeable. If

$$\mathcal{U} \subseteq \mathcal{M} \times \mathcal{R} \times \mathcal{J} \times \Gamma$$

is the literal incidence and $I = \text{pr}_{m,r} \mathcal{U}$, then the later lifted space uses the rectangular completion

$$\text{Rect}(I) = \{(m, r, j, \gamma) : (m, r) \in I, j \in \mathcal{J}, \gamma \in \Gamma\}.$$

It may contain spurious crosses. Some crosses are harmless because the terminal predicate kills them; some alter the raw parent form but not the terminal map; and some alter even the grouped physical operator. We give sharp finite-scale if-and-only-if tests for all three cases.

The second point is an exact repair. Removing the retained-pair test from the literal terminal predicate defines a coefficient-independent eligible universe and its maximal pair projection $\mathcal{I}_X^{\text{max}}$. Every declared relation selects the same native-high carrier as its effective core $\mathcal{I}_X \cap \mathcal{I}_X^{\text{max}}$. This selector identity is not an identity for a rectangular terminal operator, which still requires the terminal-faithfulness test. On the native high carrier we further separate the alignment mass W_A , which passes every non-relation terminal test, from the amount retained by $\mathcal{I}_X$. This produces an exact two-gate factorization and the correct uniform/distinguished minimax laws. Setting $\mathcal{I}_X = \mathcal{I}_X^{\text{max}}$ eliminates only relation-induced missing mass. It does not prove parent-to-terminal activation, nor may the terminal-active support be substituted retroactively for the parent support.

The third point decides what the old one-band support can and cannot do. On the large-prime band of TPC–45/47,

$$\sqrt{N_+} < p \leq N_-/R_0, \quad (2)$$

every represented atom has $r \geq R_0$. On the critical very-high face of TPC–59, $p > y \asymp Q$ implies

$$r < N_+/y \asymp J. \quad (3)$$

The exact comparison is stronger than a mere top-window localization. The balanced inherited threshold has exponent $99979/210000 > 133/400$, and every band satisfying the inherited freezing inequality remains polynomially above the critical ceiling. Its intersection with the critical face is therefore empty. If one independently retunes $R_0 \asymp J$ to create a top-window intersection, the old coefficient-blind freezing loss is $15077/105000 > 1/400$. Moreover, if the unseen lower blocks are then treated by the TPC–61 ladder and Brun–Titchmarsh certificate, their full cardinality exactly cancels the available cofactor exponent. These are genuine negative theorems for two specified certificates, not evidence that the actual weighted missing mass is large.

The resulting fork is useful. One may prove a new coefficient-specific band/freezing theorem and then seek a weighted distribution theorem for its low-cofactor complement, or canonically realize a multiscale relation and separately pay for the enlarged parent/label-reassembly gates. Either route is now falsifiable. Neither route is described as progress on the fixed- $h = 2$ parity kernel until a genuinely signed arithmetic estimate on the literal carrier has been proved.

Main results. The paper proves the following finite-scale statements.

- (i) Exact pre-terminal, terminal-atomic, and grouped-operator rectangularization criteria, including explicit spurious-cross witnesses.
- (ii) A unique least terminal-faithful pair relation when pair-only completion is possible, and a sharp criterion requiring the full incidence hypergraph when it is not.
- (iii) The maximal-eligible relation, effective-core identity, monotonicity lattice, exact alignment/completion factorization, and sharp uniform/distinguished exposure constants.
- (iv) Exact cofactor-window inclusions, the disjointness of every band certified by the inherited freezing inequality from the critical face, and the over-budget cost of the top-scale retuning.
- (v) A zero-saving theorem for the full omitted dyadic blocks under the inherited support-cardinality/harmonic certificate, even after an independently justified retuning.
- (vi) A labeled multiscale completion with zero fixed-power bookkeeping cost, together with the proof that erasing its labels is a separate reassembly map.

All exact support and operator statements are L0. Their specialization to one literal fixed shift, coefficient field, and raw atomic norm is L1. No new L2 cancellation or prime-distribution lower bound is claimed.

2 The inherited support and the missing provenance arrow

We first state exactly what is inherited and what is not. This section is not a historical survey: it fixes the logical hypotheses used in every later theorem.

2.1 The literal residual atom set

Fix one side and one direct residue cell. A source row is uniquely encoded by

$$m = d_m \ell_m \asymp Q, \quad Q = X^{267/400+o(1)}, \quad j \asymp J = X^{133/400+o(1)}. \quad (4)$$

The physical shift $h_0 \neq 0$ is fixed. On the TPC–47 residual sector, an eligible atom has the determinant relation

$$pr - mj = h_0, \quad p \in \mathcal{P}_{R_0}^\# := \{p \in \mathbb{P} : \sqrt{N_+} < p \leq N_-/R_0\}, \quad (5)$$

together with the inherited row, orbit, opposite-row, terminal, mask, squarefree, and coprimality restrictions. Write $\Lambda_X(n) := (\log n)\Phi_0(n/X)$ for $n > 0$, zero-extended to $\mathbb{Z}$, for the inherited smooth terminal weight. Thus Λ_X is not the von Mangoldt function. The literal coefficient is

$$\gamma_m^{(1)} \mathfrak{A}_{m,\gamma}^{\text{act}}(j) \mu(r) \Lambda_X(pr), \quad s = d_m r. \quad (6)$$

The large prime is unique and every occurring cofactor obeys

$$R_0 \leq r < \sqrt{N_+} \quad (7)$$

[1, 2].

Remove only the bounded row coefficient ζ_m from $\gamma_m^{(1)}$, retaining every fixed source shape and logarithmic factor. Let $c_{m,j,\gamma}$ denote the remaining coefficient-free pre-terminal factor. The structural support is defined using $c_{m,j,\gamma} \neq 0$, not using $c_{m,j,\gamma} \zeta_m \neq 0$. Thus it is fixed before the coefficient vector is chosen.

Definition 2.1 (Literal eligible and active residual incidences). Let $\mathcal{E}_X^{\text{res}}$ be the finite set of tuples $e = (m, r, p, j, \gamma)$ satisfying (5) and every declared structural support condition. Let

$$\mathcal{A}_X^{\text{res}} := \{e \in \mathcal{E}_X^{\text{res}} : c_{m,j,\gamma} \mu(r) \Lambda_X(pr) \neq 0\} \quad (8)$$

and define its pair projection

$$\mathcal{I}_X^{\text{res}} := \text{pr}_{m,r} \mathcal{A}_X^{\text{res}}. \quad (9)$$

This is a genuine coefficient-independent finite relation, but its definition is a projection of the full residual incidence. It is not a row-completion theorem and, by itself, it is not an identification with the later native high carrier.

2.2 What reassembly preserves

TPC-47 proves a coefficientwise signed Bochner reassembly for every already supported (m, r) , and TPC-48 proves an exact orthogonal r -square at fixed m before residual grouping [2, 3]. Both statements preserve the relation $\mathcal{I}_X^{\text{res}}$. They neither select a favorable pair nor add a missing one. More explicitly, at fixed m , distinct cofactors occupy the distinct residual coordinates $d_m r$, so

$$\left\| \sum_r B_{m,r} \right\|^2 = \sum_r \|B_{m,r}\|^2 \quad (10)$$

whenever the vectors are evaluated in the ungrouped residual-label space. An empty fixed-row cofactor fiber is compatible with this identity.

The orbit projector acts in the j -factor, and the all-band tiling acts on the same orbit-frequency quotient. Neither operator creates a new (m, r) -label. By contrast, the later map

$$(\gamma, d_m r, j) \mapsto (\gamma, j) \quad (11)$$

groups different residual labels and is not norm-monotone from below. Support preservation before that map is therefore not a lower reassembly estimate after it.

2.3 Where the pair relation becomes opaque

The coefficient spaces of TPC-51 and TPC-52 introduce a finite relation $\mathcal{I}_X$ described only as containing the supported pairs before coherent prime summation. TPC-53 uses the rectangular lifted space

$$\bigoplus_{(m,r) \in \mathcal{I}_X} \ell^2(\mathcal{J}_{X,a}; \ell^2(\mathcal{O}_X \times \mathcal{S}_X)), \quad (12)$$

and its exact factor dictionary is valid for any declared finite retained set. TPC-54 then keeps only the multiplicity

$$N_r(m) = \#\{r : (m, r) \in \mathcal{I}_X\} \quad (13)$$

and explicitly leaves balanced occupancy unproved [4–7].

The displayed interfaces therefore do not prove any of

$$\mathcal{I}_m \cap \text{supp } \nu_m \neq \emptyset \quad \text{on every native-high active row,} \quad (14)$$

$$\inf_m \frac{\nu_m(\mathcal{I}_m)}{\nu_m(\mathbb{N})} > 0, \quad (15)$$

$$\sup_m \frac{\nu_m(\mathcal{I}_m^c)}{\nu_m(\mathbb{N})} \leq X^{-\sigma+o(1)} \quad \text{for some } \sigma > 0. \quad (16)$$

The statement is a logical non-implication. It does not assert that the actual arithmetic relation has a bad row.

Proposition 2.2 (Support-preservation boundary). *Under the displayed TPC-47/48 interface, $\mathcal{I}_X^{\text{res}}$ is finite, is preserved by coefficientwise signed reassembly and orbit-band tiling, and has the fixed-row orthogonality (10). Those statements do not imply any of (14)–(16) for an independently specified native-high measure family (ν_m) .*

Proof. Finiteness follows from the finite row, orbit, terminal, and opposite-row supports. The residual label is $d_m r$; at fixed m , its injectivity in r gives (10). The coefficientwise identities in TPC-47/48 are indexed by the already supported pair and hence do not alter its projection. For the non-implication, take one source row, let $\mathcal{I}_m^{\text{res}} = \{r_0\}$, and let an otherwise admissible native-high measure be $\nu_m = \delta_{r_1}$ with $r_1 \neq r_0$. Every stated finite-support and orthogonality property remains true, while $\nu_m(\mathcal{I}_m^{\text{res}}) = 0 < \nu_m(\mathbb{N})$. This model witnesses insufficiency of the displayed hypotheses only. $\square$

The next sections replace the opaque arrow by a full-incidence calculus. That calculus will show exactly when a pair projection is faithful and exactly which extra theorem would be needed to identify it with the native high support.

3 A finite full-incidence calculus

The retained-pair question is finite at every value of X . We therefore begin with a finite-dimensional theorem which keeps all four pre-terminal labels. The theorem distinguishes preservation of the pre-terminal raw form, preservation of the terminal atomic form, and preservation of the grouped physical operator. These notions coincide only under additional incidence hypotheses.

3.1 The ambient incidence and its pair projection

Let $\mathcal{M}, \mathcal{R} \subset \mathbb{N}_{\geq 1}$ and $\mathcal{J} \subset \mathbb{Z}$ be finite, and let Γ be a finite label set. Put

$$\Omega := \mathcal{M} \times \mathcal{R} \times \mathcal{J} \times \Gamma \quad (17)$$

and write

$$u = (m, r, j, \gamma), \quad m(u) = m,$$

and take the row-coefficient space to be $Z := \mathbb{C}^{\mathcal{M}}$. All sets, structural coefficients, and maps below are fixed before $z \in Z$ is chosen. A version with a finite base fiber $B_m \subseteq \mathcal{J} \times \Gamma$ depending on m is obtained by replacing $\mathcal{J} \times \Gamma$ by B_m ; none of the proofs changes.

Let $c = (c_u)_{u \in \Omega}$ be the coefficient-free structural field. It contains every literal source shape, mask, opposite-row factor, and orbit factor preceding terminalization, but not the free row coefficient z_m . For $V \subseteq \Omega$, define the zero-extended synthesis operator

$$(C_V z)_u := \mathbf{1}_V(u) c_u z_{m(u)}, \quad C_V : Z \longrightarrow \ell^2(\Omega). \quad (18)$$

Thus every carrier is compared in the same counting-measure Hilbert space.

Definition 3.1 (Full incidence, pair projection, and rectangle). Let $\mathcal{U} \subseteq \Omega$ be a declared full pre-terminal incidence. Its pair projection and rectangular completion are

$$I(\mathcal{U}) := \text{pr}_{m,r} \mathcal{U} = \{(m, r) : \text{there exists } (j, \gamma) \text{ with } (m, r, j, \gamma) \in \mathcal{U}\}, \quad (19)$$

$$\text{Rect}(I) := \{(m, r, j, \gamma) \in \Omega : (m, r) \in I\}. \quad (20)$$

The associated spurious-cross set is

$$\mathcal{S}(\mathcal{U}) := \text{Rect}(I(\mathcal{U})) \setminus \mathcal{U}. \quad (21)$$

By construction, $\mathcal{U} \subseteq \text{Rect}(I(\mathcal{U}))$. This containment does not say that the new crossed coordinates in $\mathcal{S}(\mathcal{U})$ have zero structural coefficient or fail the terminal test.

3.2 Terminal restriction and the grouped operator

Fix $h \in \mathbb{Z}$. For $u = (m, r, j, \gamma)$, put

$$p_h(u) := \frac{mj + h}{r} \quad (22)$$

when the quotient is a positive integer. Let $\mathbf{1}_{\text{term},h}(u)$ be zero when that quotient is not a positive integer, and otherwise the indicator of the complete declared terminal predicate: the quotient in (22) is a prime in the stated support, and every required terminal cutoff, squarefree condition, coprimality condition, literal mask, and nonvanishing terminal profile holds. Define

$$t_h(u) := \mu(r) \Lambda_X(mj + h) \mathbf{1}_{\text{term},h}(u), \quad (D_h x)_u := t_h(u) x_u. \quad (23)$$

Here $\mu(r)$ is the single inherited cofactor sign, and $\Lambda_X(n) = (\log n) \Phi_0(n/X)$ for $n > 0$, zero-extended to $\mathbb{Z}$, is the inherited smooth terminal weight rather than the von Mangoldt function. No additional Möbius factor may be inserted in the physical specialization.

Let $\mathcal{Q}$ be a finite output-key set, let $q : \Omega \rightarrow \mathcal{Q}$ be fixed, and define

$$(Gx)_a := \sum_{\substack{u \in \Omega \\ q(u)=a}} x_u, \quad G : \ell^2(\Omega) \longrightarrow \ell^2(\mathcal{Q}). \quad (24)$$

For the fixed-shift residual packet the literal key is

$$q(m, r, j, \gamma) = (\gamma, d_m r, j). \quad (25)$$

For $V \subseteq \Omega$, write

$$P_V(z) := \|C_V z\|_2^2, \quad (26)$$

$$A_{h,V}(z) := \|D_h C_V z\|_2^2, \quad (27)$$

$$F_{h,V} := G D_h C_V. \quad (28)$$

The first two forms use raw counting measure. The last map is coherent on each output fiber, so its squared norm need not be a sum of atomic squares.

3.3 Exact rectangularization identities

Theorem 3.2 (Rectangularization difference identities). *Let $\mathcal{U} \subseteq V \subseteq \Omega$, and put $\mathcal{S} := V \setminus \mathcal{U}$. Then*

$$C_V - C_{\mathcal{U}} = C_{\mathcal{S}}, \quad (29)$$

$$D_h C_V - D_h C_{\mathcal{U}} = D_h C_{\mathcal{S}}, \quad (30)$$

$$F_{h,V} - F_{h,\mathcal{U}} = G D_h C_{\mathcal{S}}. \quad (31)$$

Moreover, for every $z \in Z$,

$$P_V(z) - P_{\mathcal{U}}(z) = \sum_{u \in \mathcal{S}} |c_u|^2 |z_{m(u)}|^2, \quad (32)$$

$$A_{h,V}(z) - A_{h,\mathcal{U}}(z) = \sum_{u \in \mathcal{S}} |t_h(u)c_u|^2 |z_{m(u)}|^2. \quad (33)$$

Finally, the (a, m) -matrix entry of the grouped difference is

$$\boxed{(F_{h,V} - F_{h,\mathcal{U}})_{a,m} = \sum_{\substack{u \in \mathcal{S} \\ q(u)=a, \ m(u)=m}} t_h(u)c_u.} \quad (34)$$

In particular, the identities apply to $V = \text{Rect}(I(\mathcal{U}))$ and $\mathcal{S} = \mathcal{S}(\mathcal{U})$.

Proof. Since $V = \mathcal{U} \dot{\cup} \mathcal{S}$, one has $\mathbf{1}_V - \mathbf{1}_{\mathcal{U}} = \mathbf{1}_{\mathcal{S}}$. Substitution in (18) proves (29); applying D_h and G gives the next two identities. The vectors on $\mathcal{U}$ and $\mathcal{S}$ have disjoint coordinate supports, before and after the diagonal map D_h , so Pythagoras proves the two form identities. Finally, evaluate (31) at the row basis vector e_m and read its a -coordinate. $\square$

The positive identities (32) and (33) must not be transferred through G : terms in (34) can cancel.

Theorem 3.3 (Three exact fidelity criteria). *Let $I = I(\mathcal{U})$, and put $\mathcal{S} = \text{Rect}(I) \setminus \mathcal{U}$. Then:*

- (i) *The rectangular lift preserves the pre-terminal synthesis and raw form exactly if and only if*

$$\begin{aligned} C_{\text{Rect}(I)} = C_{\mathcal{U}} &\iff P_{\text{Rect}(I)} = P_{\mathcal{U}} \text{ on } Z \\ &\iff c_u = 0 \quad (u \in \mathcal{S}). \end{aligned} \quad (35)$$

- (ii) *It preserves the terminal atomic synthesis and raw form exactly if and only if*

$$\begin{aligned} D_h C_{\text{Rect}(I)} = D_h C_{\mathcal{U}} &\iff A_{h,\text{Rect}(I)} = A_{h,\mathcal{U}} \text{ on } Z \\ &\iff t_h(u)c_u = 0 \quad (u \in \mathcal{S}). \end{aligned} \quad (36)$$

- (iii) *It preserves the grouped physical operator exactly if and only if*

$$\boxed{F_{h,\text{Rect}(I)} = F_{h,\mathcal{U}} \iff \sum_{\substack{u \in \mathcal{S} \\ q(u)=a, \ m(u)=m}} t_h(u)c_u = 0 \text{ for every } (a, m).} \quad (37)$$

Consequently, pre-terminal fidelity implies terminal-atomic fidelity, which implies grouped-operator fidelity. Neither converse holds for a general grouping map.

Proof. Map equality implies equality of the corresponding squared norms. Conversely, take $z = e_m$ in (32). Equality of the pre-terminal forms gives

$$\sum_{\substack{u \in \mathcal{S} \\ m(u)=m}} |c_u|^2 = 0 \quad (m \in \mathcal{M}),$$

so every c_u on $\mathcal{S}$ vanishes. The terminal argument is identical using (33). The grouped criterion is (34), because a linear map from $\mathbb{C}^{\mathcal{M}}$ is zero exactly when every matrix entry is zero. $\square$

Example 3.4 (The hierarchy is strict). *If a spurious coordinate u_0 has $c_{u_0} \neq 0$ but $t_h(u_0) = 0$, pre-terminal fidelity fails while both later fidelities can hold. Separately, two spurious atoms in the same row and output fiber with terminal coefficients 1 and -1 violate terminal-atomic fidelity but cancel in the grouped operator. The latter is an abstract example; the literal key (25) has the singleton property below.*

3.4 Spurious-cross witnesses

Proposition 3.5 (Exact spurious-cross witnesses). *Let $I = I(\mathcal{U})$, let $\mathcal{S} = \text{Rect}(I) \setminus \mathcal{U}$, and take $u_0 \in \mathcal{S}$, with $m_0 = m(u_0)$ and $a_0 = q(u_0)$.*

(a) *If $c_{u_0} \neq 0$, then*

$$P_{\text{Rect}(I)}(e_{m_0}) - P_{\mathcal{U}}(e_{m_0}) \geq |c_{u_0}|^2 > 0. \quad (38)$$

(b) *If $t_h(u_0)c_{u_0} \neq 0$, then*

$$A_{h,\text{Rect}(I)}(e_{m_0}) - A_{h,\mathcal{U}}(e_{m_0}) \geq |t_h(u_0)c_{u_0}|^2 > 0. \quad (39)$$

(c) *If*

$$\Sigma_{a_0,m_0}(\mathcal{S}) := \sum_{\substack{u \in \mathcal{S} \\ q(u)=a_0, m(u)=m_0}} t_h(u)c_u \neq 0, \quad (40)$$

then

$$\|(F_{h,\text{Rect}(I)} - F_{h,\mathcal{U}})e_{m_0}\|_2 \geq |\Sigma_{a_0,m_0}(\mathcal{S})| > 0. \quad (41)$$

In particular, this holds when u_0 is the only terminal-active spurious atom in its (a_0, m_0) -fiber.

Proof. The first two assertions are the single-summand lower bounds in (32) and (33). For the last assertion, the a_0 -coordinate of the displayed difference vector is exactly $\Sigma_{a_0,m_0}(\mathcal{S})$. $\square$

Corollary 3.6 (The literal residual key removes accidental rowwise cancellation). *Assume every row integer m determines d_m , and use $q(u) = (\gamma, d_m r, j)$. For fixed (a, m) , at most one $u \in \Omega$ satisfies $q(u) = a$ and $m(u) = m$. Hence*

$$F_{h,\text{Rect}(I)} = F_{h,\mathcal{U}} \iff D_h C_{\text{Rect}(I)} = D_h C_{\mathcal{U}}. \quad (42)$$

Cancellation between different rows cannot repair this uniform equality, because the row variables z_m are independent.

Proof. Fix $a = (\gamma, s, j)$ and m . The row determines d_m , and $s = d_m r$ determines r . Thus m, r, j, γ , hence u , are fixed. Every sum in (37) has at most one term. $\square$

This corollary does not identify the inherited relation $\mathcal{I}_X$ with the projection of any particular full incidence. It says only that, after the literal full incidence has been written down, an active spurious cross cannot be hidden by the residual-key grouping $q = (\gamma, d_m r, j)$ in a coefficient-uniform comparison. A later map which erases $d_m r$ can merge distinct cofactors and requires the separate fiber/kernel test of section 7.

4 Canonical faithful completion and the hypergraph alternative

The preceding calculus decides whether one proposed pair projection is faithful. We now solve the inverse problem: given the literal full incidence, when does any pair-only rectangular representation exist, and what is its unique least relation? The answer depends on whether one asks to preserve the parent form or only the terminal atomic field.

4.1 Two canonical active incidences

Continue with the finite data of [section 3](#). Because the structural support is fixed before z , the following sets are also coefficient-independent.

Definition 4.1 (Pre-terminal and terminal active incidences). For a full incidence $\mathcal{U} \subseteq \Omega$, define

$$\mathcal{U}^{\text{pre},+} := \{u \in \mathcal{U} : c_u \neq 0\}, \quad (43)$$

$$\mathcal{U}_h^+ := \{u \in \mathcal{U} : t_h(u)c_u \neq 0\}. \quad (44)$$

The second set is contained in the first. It may depend on the fixed physical shift and terminal support, but not on the row coefficient vector. It must not be substituted for $\mathcal{U}^{\text{pre},+}$ in an upstream parent form: that substitution would make terminal activation true by definition.

Proposition 4.2 (Unique minimal coordinate supports). *Let $V \subseteq \mathcal{U}$.*

(i) *The following are equivalent:*

$$C_V = C_{\mathcal{U}}, \quad P_V = P_{\mathcal{U}} \text{ on } Z, \quad \mathcal{U}^{\text{pre},+} \subseteq V. \quad (45)$$

Consequently $\mathcal{U}^{\text{pre},+}$ is the unique least coordinate support preserving the pre-terminal synthesis and form.

(ii) *The following are equivalent:*

$$D_h C_V = D_h C_{\mathcal{U}}, \quad A_{h,V} = A_{h,\mathcal{U}} \text{ on } Z, \quad \mathcal{U}_h^+ \subseteq V. \quad (46)$$

Consequently $\mathcal{U}_h^+$ is the unique least coordinate support preserving the terminal atomic synthesis and form.

Proof. If V contains the stated active set, every omitted coordinate has zero coefficient in the relevant synthesis. Conversely, if $u_0 \in \mathcal{U}^{\text{pre},+} \setminus V$, then, at $z = e_{m(u_0)}$,

$$P_{\mathcal{U}}(z) - P_V(z) \geq |c_{u_0}|^2 > 0.$$

The terminal statement follows identically with $|t_h(u_0)c_{u_0}|^2$. Thus every preserving support contains the corresponding active set. $\square$

There need not be a unique least coordinate support preserving only the grouped operator. Exact cancellations inside a grouped fiber can make different subsets synthesize the same output. The terminal atomic support is canonical because it is defined before that cancellation.

4.2 Active rectangular closure

For $I \subseteq \mathcal{M} \times \mathcal{R}$, define

$$\text{PRect}(I) := \{u \in \text{Rect}(I) : c_u \neq 0\}, \quad (47)$$

$$\text{TRect}_h(I) := \{u \in \text{Rect}(I) : t_h(u)c_u \neq 0\}. \quad (48)$$

Both operations are monotone in I .

Definition 4.3 (Atomic-faithful pair relation). A pair relation I is

(a) *pre-terminal faithful* for $\mathcal{U}$ if

$$\text{PRect}(I) = \mathcal{U}^{\text{pre},+}; \quad (49)$$

(b) *terminal-atomic faithful* for $(\mathcal{U}, h)$ if

$$\text{TRect}_h(I) = \mathcal{U}_h^+. \quad (50)$$

These definitions demand both completeness and absence of active spurious crosses. They are stronger than equality after grouping and are the appropriate notions for raw exposure forms.

Theorem 4.4 (Unique least terminal-faithful pair relation). *Put*

$$I_h^{\min} := \text{pr}_{m,r} \mathcal{U}_h^+. \quad (51)$$

There exists a terminal-atomic faithful pair relation for $(\mathcal{U}, h)$ if and only if

$$\boxed{\text{TRect}_h(I_h^{\min}) = \mathcal{U}_h^+}. \quad (52)$$

When this condition holds, $I_h^{\min}$ is the unique least faithful relation. More precisely, I is terminal-atomic faithful if and only if

$$I_h^{\min} \subseteq I \quad \text{and} \quad \text{TRect}_h(I \setminus I_h^{\min}) = \emptyset. \quad (53)$$

Proof. Every $u \in \mathcal{U}_h^+$ has its pair in $I_h^{\min}$, hence

$$\mathcal{U}_h^+ \subseteq \text{TRect}_h(I_h^{\min}). \quad (54)$$

If a faithful relation I exists, it contains the pair of every $u \in \mathcal{U}_h^+$, so $I_h^{\min} \subseteq I$. Monotonicity gives

$$\mathcal{U}_h^+ \subseteq \text{TRect}_h(I_h^{\min}) \subseteq \text{TRect}_h(I) = \mathcal{U}_h^+,$$

which proves (52). Conversely, that condition says exactly that $I_h^{\min}$ is faithful, and every faithful relation contains it.

Assume the criterion holds. If I is faithful, any active atom created by a pair in $I \setminus I_h^{\min}$ has a pair absent from the projection of $\mathcal{U}_h^+$, and so lies outside $\mathcal{U}_h^+$, a contradiction. This proves necessity of the second condition in (53). The two conditions are sufficient because the first retains all of $\text{TRect}_h(I_h^{\min}) = \mathcal{U}_h^+$, while the second adds no active atom. $\square$

Corollary 4.5 (Unique least pre-terminal faithful relation). *Put*

$$I_{\text{pre}}^{\min} := \text{pr}_{m,r} \mathcal{U}^{\text{pre},+}. \quad (55)$$

A pre-terminal faithful pair relation exists if and only if

$$\boxed{\text{PRect}(I_{\text{pre}}^{\min}) = \mathcal{U}^{\text{pre},+}}. \quad (56)$$

When it exists, $I_{\text{pre}}^{\min}$ is the unique least one.

Proof. Repeat the proof of [theorem 4.4](#), replacing TRect_h by PRect and $\mathcal{U}_h^+$ by $\mathcal{U}^{\text{pre},+}$. $\square$

Condition (56) is stronger than (52). A crossed coordinate can carry nonzero parent energy and still be terminal-null. Thus exact terminal factorization through a pair relation does not justify using the rectangular lift's parent form as the upstream parent form.

4.3 Pair projection is not full provenance

Example 4.6 (The same pair projection supports different incidences). *Take one row m , two cofactors r_1, r_2 , and two base labels $b_1, b_2 \in \mathcal{J} \times \Gamma$. In the resulting four-point rectangle let*

$$\mathcal{U}_{\text{diag}} := \{(m, r_1, b_1), (m, r_2, b_2)\}, \quad (57)$$

$$\mathcal{U}_{\text{cross}} := \{(m, r_1, b_2), (m, r_2, b_1)\}, \quad (58)$$

$$\mathcal{U}_{\text{full}} := \{m\} \times \{r_1, r_2\} \times \{b_1, b_2\}. \quad (59)$$

All three pair projections equal $\{(m, r_1), (m, r_2)\}$. Let every point of the ambient rectangle be terminal-active and let the output key resolve individual atoms. The projected relation is faithful for $\mathcal{U}_{\text{full}}$, but no pair relation is terminal-atomic faithful for either $\mathcal{U}_{\text{diag}}$ or $\mathcal{U}_{\text{cross}}$: retaining both actual pairs necessarily creates the other two active crosses.

The atom-resolved operators are different even if all coefficients have modulus one. To make the scalar raw forms differ as well, assign terminal coefficient modulus one to the diagonal points and modulus two to the cross points. At $z = e_m$, the diagonal and cross forms are then 2 and 8, while their pair projections remain equal.

Example 4.7 (Grouped equality can conceal a nonfaithful rectangle). *Suppose $\text{Rect}(I_h^{\min}) \setminus \mathcal{U}_h^+$ contains two active crosses with the same row and grouped output key, having terminal coefficients 1 and -1 . The rectangular lift is not terminal-atomic faithful, yet the spurious columns cancel in the grouped operator. Hence grouped equality does not certify an exposure identity. Under $q = (\gamma, d_m r, j)$, this cancellation is excluded by [corollary 3.6](#).*

These are finite-incidence countermodels, not asserted arithmetic counterexamples inside the actual TPC carrier. Their exact conclusion is that hypotheses recording only the pair projection cannot reconstruct the full incidence.

4.4 When the full incidence hypergraph is mandatory

Theorem 4.8 (Sharp hypergraph-necessity criterion). *If*

$$\text{TRect}_h(I_h^{\min}) \setminus \mathcal{U}_h^+ \neq \emptyset, \quad (60)$$

then no pair relation is simultaneously complete for every atom of $\mathcal{U}_h^+$ and free of terminal-active spurious coordinates. The full incidence hypergraph

$$\mathcal{U}_h^+ \subseteq \mathcal{M} \times \mathcal{R} \times \mathcal{J} \times \Gamma$$

is then the unique least terminal-atomic coordinate subcarrier among subsets of the literal carrier $\mathcal{U}$.

If instead the terminal criterion (52) holds but the pre-terminal criterion (56) fails, a pair relation can represent the terminal atomic field exactly but does not preserve the parent raw form. A proof chain using that parent must retain the full pre-terminal incidence or insert a separate proved parent comparison.

Proof. The first assertion follows from [theorem 4.4](#); unique coordinate minimality follows from [proposition 4.2](#). The second assertion is the simultaneous application of [theorem 4.4](#) and [corollary 4.5](#), together with the positive discrepancy (32). $\square$

The term *hypergraph* records the four-label incidence; it introduces no new averaging device. Keeping it merely refuses an identification which the pair projection cannot reconstruct.

4.5 The noncircular fixed-shift audit

Let $\mathcal{U}_X^{\text{lit}}$ be obtained by copying every pre-terminal coordinate of the upstream coefficientwise field, including its row, cofactor, orbit, and opposite-row support tests, and put

$$I_X^{\text{lit}} := \text{pr}_{m,r} \mathcal{U}_X^{\text{lit}}. \quad (61)$$

This projection alone does not prove that a later rectangular space is faithful.

Proposition 4.9 (Noncircular provenance criterion). *Identification of a later retained relation $\mathcal{I}_X$ with the literal pair projection is valid at the following respective levels only after the corresponding statements have been proved:*

$$\mathcal{I}_X = I_X^{\text{lit}} \quad \text{and} \quad \text{PRect}(I_X^{\text{lit}}) = (\mathcal{U}_X^{\text{lit}})^{\text{pre},+}, \quad (62)$$

$$\mathcal{I}_X = I_{X,h}^{\text{min}} \quad \text{and} \quad \text{TRect}_h(I_{X,h}^{\text{min}}) = (\mathcal{U}_X^{\text{lit}})_h^+, \quad (63)$$

where

$$I_{X,h}^{\text{min}} := \text{pr}_{m,r} (\mathcal{U}_X^{\text{lit}})_h^+.$$

The first line proves pre-terminal and hence terminal fidelity. The second proves terminal-atomic fidelity only. Finiteness of $\mathcal{I}_X$, equality of cardinalities, or agreement of dyadic cofactor ranges does not imply either line.

Proof. Sufficiency and the exact logical strength of the two lines are [corollary 4.5](#) and [theorem 4.4](#). Conversely, a missing actual pair deletes a nonzero coordinate, while failed rectangularity adds an active spurious cross. Cardinality and range data record neither event, as [example 4.6](#) demonstrates. $\square$

Remark 4.10 (No inherited identification is assumed). Neither (62) nor (63) is inferred from the phrase *supported pairs*. In the physical application one must return to the TPC-47 coefficient field, write $\mathcal{U}_X^{\text{lit}}$ by a literal set-builder formula, and verify the chosen equality atom by atom. If that verification fails, the correct repair is the full-incidence carrier, not a redefinition of $\mathcal{I}_X$ chosen to make exposure favorable.

On the very-high face $p > \sqrt{N_+}$, largest-prime factorization makes the terminal cofactor unique for fixed (m, j, γ) . This helps check (63), but does not prove the required saturation: one must still show that every base label crossed with a retained pair is present or terminal-null.

5 Maximal eligibility and exact row exposure

The incidence calculus separates two questions which were previously hidden inside the phrase “retained pair.” First, does a native high atom pass all terminal tests other than membership in the pair relation? Second, if it does, is its recovered pair retained? This section disintegrates the two questions before any residual or native grouping. All sets are fixed before the row coefficient is chosen.

5.1 The relation-free admissible high carrier

Let $\mathcal{W}_H(y)$ be the complete nonzero native very-high carrier on one fixed side and direct residue slice. Every $w \in \mathcal{W}_H(y)$ canonically recovers

$$m(w), \quad j(w), \quad n(w) = m(w)j(w) + h_0, \quad p(w) = P^+(n(w)) > y, \quad r(w) = \frac{n(w)}{p(w)}. \quad (64)$$

Write its literal amplitude as

$$b_w(\zeta) = \beta_w \zeta_{m(w)}, \quad (65)$$

where β_w contains every coefficient-free source, mask, orbit, opposite-row, and shaped terminal factor. No free atom phase is inserted.

Remove only the retained-pair membership test from the complete terminal admission predicate. Let

$$A_X(w) \in \{0, 1\} \quad (66)$$

be the conjunction of all remaining declared tests: terminal-prime and orbit scope, nonvanishing terminal weight, squarefree and coprimality conditions, literal mask and profile support, and every further terminal-only condition. Define

$$\mathcal{W}_A(y) := \{w \in \mathcal{W}_H(y) : A_X(w) = 1\}, \quad (67)$$

$$\mathcal{I}_X^{\max} := \{(m(w), r(w)) : w \in \mathcal{W}_A(y)\}. \quad (68)$$

The superscript “max” refers only to removal of the retained-pair test. It does not enlarge any other literal support. Under the exact terminal–Walsh coordinate injection, $\mathcal{I}_X^{\max}$ is the pair projection of the coefficient-free terminal incidence. Identifying it with the residual projection $\mathcal{I}_X^{\text{res}}$ from [definition 2.1](#) additionally requires equality of the two declared support systems; this equality is not inferred from their common scales.

Let $\mathcal{I}_X \subseteq \mathcal{M}_X \times \mathbb{N}$ be any declared finite, coefficient-independent pair relation and put

$$\mathcal{I}_m := \{r : (m, r) \in \mathcal{I}_X\}, \quad \mathcal{I}_X^{\text{eff}} := \mathcal{I}_X \cap \mathcal{I}_X^{\max}. \quad (69)$$

Its represented high carrier and its full missing complement are

$$\mathcal{W}_R^{\mathcal{I}} := \{w \in \mathcal{W}_A(y) : (m(w), r(w)) \in \mathcal{I}_X\}, \quad (70)$$

$$\mathcal{W}_M^{\mathcal{I}} := \mathcal{W}_H(y) \setminus \mathcal{W}_R^{\mathcal{I}}. \quad (71)$$

Thus $\mathcal{W}_M^{\mathcal{I}}$ includes both non-relation failure and failure of a different terminal test. In the identical-support specialization of TPC–60/61, $A_X = 1$ on $\mathcal{W}_H$, so only the first mechanism remains [\[9, 10\]](#).

Proposition 5.1 (Effective core and the eligibility lattice). *For every declared relation $\mathcal{I}_X$,*

$$\boxed{\mathcal{W}_R^{\mathcal{I}} = \mathcal{W}_R^{\mathcal{I}_X^{\text{eff}}}}. \quad (72)$$

If $\mathcal{I}_X \subseteq \mathcal{J}_X$, then

$$\mathcal{W}_R^{\mathcal{I}} \subseteq \mathcal{W}_R^{\mathcal{J}}. \quad (73)$$

Moreover, $\mathcal{I}_X^{\max}$ is the unique least pair relation which represents every atom of $\mathcal{W}_A(y)$. Any superset of $\mathcal{I}_X^{\max}$ gives the same represented carrier $\mathcal{W}_A(y)$.

Proof. Membership in $\mathcal{W}_R^{\mathcal{I}}$ requires both $A_X(w) = 1$ and $(m(w), r(w)) \in \mathcal{I}_X$. The first condition already places the recovered pair in $\mathcal{I}_X^{\max}$, proving (72). Relation inclusion gives (73). Finally, every relation which represents all of $\mathcal{W}_A$ must contain the pair of every one of its atoms, hence must contain their projection $\mathcal{I}_X^{\max}$. Conversely, that projection represents every admissible atom by definition. $\square$

The proposition is a maximal-support identity. It is not a statement that $\mathcal{W}_A$ has positive mass relative to $\mathcal{W}_H$, and it is not a theorem that the inherited relation equals $\mathcal{I}_X^{\max}$.

5.2 Two realized measures and the exact factorization

For a source row m , define finite positive measures on $\mathbb{N}$ by

$$\nu_m^H(r) := \sum_{\substack{w \in \mathcal{W}_H(y) \\ m(w)=m, r(w)=r}} |\beta_w|^2, \quad (74)$$

$$\nu_m^A(r) := \sum_{\substack{w \in \mathcal{W}_A(y) \\ m(w)=m, r(w)=r}} |\beta_w|^2. \quad (75)$$

Put

$$W_H(m) := \nu_m^H(\mathbb{N}), \quad W_A(m) := \nu_m^A(\mathbb{N}), \quad (76)$$

$$W_R^{\mathcal{I}}(m) := \nu_m^A(\mathcal{I}_m), \quad W_M^{\mathcal{I}}(m) := W_H(m) - W_R^{\mathcal{I}}(m). \quad (77)$$

All weights use raw counting-measure atoms. In particular, neither ν_m^H nor ν_m^A is divided by a row, orbit, cofactor, or native fiber multiplicity.

For every row with $W_H(m) > 0$, define the alignment factor

$$\alpha_m := \frac{W_A(m)}{W_H(m)} \in [0, 1]. \quad (78)$$

When $W_A(m) > 0$, define the relation-completion factor

$$\theta_m(\mathcal{I}) := \frac{\nu_m^A(\mathcal{I}_m)}{W_A(m)} \in [0, 1]; \quad (79)$$

when $W_A(m) = 0$, put $\theta_m(\mathcal{I}) := 0$.

Theorem 5.2 (Exact alignment–completion factorization). *For every native-high active row,*

$$\boxed{\frac{W_R^{\mathcal{I}}(m)}{W_H(m)} = \alpha_m \theta_m(\mathcal{I})}. \quad (80)$$

Consequently,

$$W_R^{\mathcal{I}}(m) > 0 \iff W_A(m) > 0 \text{ and } \mathcal{I}_m \cap \text{supp } \nu_m^A \neq \emptyset. \quad (81)$$

For the canonical maximal relation,

$$\theta_m(\mathcal{I}_X^{\max}) = 1 \text{ whenever } W_A(m) > 0, \quad \frac{W_R^{\max}(m)}{W_H(m)} = \alpha_m. \quad (82)$$

Thus maximal relation completion removes only relation-induced missing mass; it does not remove the alignment defect $1 - \alpha_m$.

Proof. If $W_A(m) > 0$, substitute $W_R^{\mathcal{I}}(m) = \nu_m^A(\mathcal{I}_m)$ and insert $W_A(m)$ between numerator and denominator. If $W_A(m) = 0$, both sides of (80) vanish by convention. A positive finite measure charges $\mathcal{I}_m$ precisely when that set meets its support, proving (81). Finally, $\mathcal{I}_X^{\max}$ contains every pair in $\text{supp } \nu_m^A$, which gives (82). $\square$

Under identical support one has $\nu_m^A = \nu_m^H$ and $\alpha_m = 1$. Then (80) reduces exactly to the TPC–61 realized-cofactor formula

$$\frac{W_R^{\mathcal{I}}(m)}{W_H(m)} = \frac{\nu_m^H(\mathcal{I}_m)}{\nu_m^H(\mathbb{N})}. \quad (83)$$

The specialization is useful only after identical support itself has been proved for the declared packet.

5.3 Sharp uniform and distinguished constants

For a row-coefficient vector ζ , let

$$\mathcal{D}_H(\zeta) := \sum_m W_H(m) |\zeta_m|^2, \quad (84)$$

$$\mathcal{D}_R^{\mathcal{I}}(\zeta) := \sum_m W_R^{\mathcal{I}}(m) |\zeta_m|^2, \quad \mathcal{D}_M^{\mathcal{I}}(\zeta) := \mathcal{D}_H(\zeta) - \mathcal{D}_R^{\mathcal{I}}(\zeta). \quad (85)$$

Theorem 5.3 (Uniform and distinguished exposure minimax). *Assume the declared uniform coefficient class contains a nonzero multiple of each active row direction. If $\mathcal{D}_H$ is not identically zero, then*

$$\inf_{\mathcal{D}_H(\zeta) > 0} \frac{\mathcal{D}_R^{\mathcal{I}}(\zeta)}{\mathcal{D}_H(\zeta)} = \min_{m: W_H(m) > 0} \alpha_m \theta_m(\mathcal{I}), \quad (86)$$

$$\sup_{\mathcal{D}_H(\zeta) > 0} \frac{\mathcal{D}_M^{\mathcal{I}}(\zeta)}{\mathcal{D}_H(\zeta)} = \max_{m: W_H(m) > 0} (1 - \alpha_m \theta_m(\mathcal{I})). \quad (87)$$

For one distinguished physical vector ζ_* with $\mathcal{D}_H(\zeta_*) > 0$, the exact represented fraction is instead

$$\rho_*(X) := \frac{\mathcal{D}_R^{\mathcal{I}}(\zeta_*)}{\mathcal{D}_H(\zeta_*)} = \sum_{m: W_H(m) > 0} \frac{W_H(m) |\zeta_{*,m}|^2}{\mathcal{D}_H(\zeta_*)} \alpha_m \theta_m(\mathcal{I}). \quad (88)$$

Proof. By [theorem 5.2](#), each quotient is a convex combination of the row values $\alpha_m \theta_m(\mathcal{I})$, with weights $W_H(m) |\zeta_m|^2 / \mathcal{D}_H(\zeta)$. It is bounded below by their minimum, and concentration on a minimizing coordinate attains that value after a harmless scalar rescaling into the declared class. Apply the same argument to $1 - \alpha_m \theta_m$ for the missing quotient. Keeping the literal weights of ζ_* gives [\(88\)](#). $\square$

At one finite scale, coefficient-uniform positive exposure is therefore equivalent to

$$\min_{m: W_H(m) > 0} \alpha_m \theta_m(\mathcal{I}) > 0. \quad (89)$$

A scale-uniform constant requires the infimum of this minimum over all large X to be positive. The much stronger fixed-power missing-mass target requires

$$\max_{m: W_H(m) > 0} (1 - \alpha_m \theta_m(\mathcal{I})) \leq X^{-\sigma + o(1)} \quad (90)$$

for some $\sigma > 0$. Mere rowwise nonemptiness proves neither statement.

5.4 The sharp support no-go and its scope

Corollary 5.4 (Finite-relation insufficiency). *The inherited hypotheses “ $\mathcal{I}_X$ is finite and fixed before ζ ” do not imply a positive value in [\(86\)](#), even under identical support. They also do not imply [\(90\)](#).*

Proof. The stated hypotheses permit an otherwise active row m_0 for which $\mathcal{I}_{m_0} \cap \text{supp } \nu_{m_0}^A = \emptyset$. Under identical support, $\alpha_{m_0} = 1$, but $\theta_{m_0} = 0$. Hence the exact minimum is zero and the exact maximum missing fraction is one. This is an admissible model of the displayed interface, not an assertion that the literal upstream construction necessarily chooses such a row. $\square$

The distinguished route is logically different. It may survive a bad row when $\zeta_{*,m} = 0$ there, but the upper bound $|\zeta_{*,m}| \ll 1$ alone supplies no lower bound for [\(88\)](#). Likewise, defining $\mathcal{I}_X^{\max}$

from support gives the exact value $\theta_m = 1$, but does not prove $\alpha_m > 0$, parent-to-high activation, or any post-grouping lower bound.

The lattice and factorization theorems are L0 finite-incidence statements. Their use with (64), the literal β_w , one fixed h_0 , and the raw native atomic norm is an L1 physical-interface specialization. The support no-go closes only an inference from opaque finite relations. It neither lower-bounds actual missing mass nor estimates the signed parity kernel, and it has no L2 prime-pair consequence.

6 The single-band completion barrier

We now compare the literal global prime band inherited by TPC-47 with the critical very-high face of TPC-59. The comparison is made at the same fixed shift and on the same target interval. It has two parts. First, an exact constant-factor sandwich identifies which cofactors one band can see. Second, the TPC-61 harmonic certificate is evaluated on every full dyadic block that the band omits. Neither part estimates the actual mass of those blocks.

6.1 An exact constant-factor window

Let every target satisfy

$$N_- \leq n = mj + h_0 = pr \leq N_+, \quad \kappa_N := \frac{N_+}{N_-}. \quad (91)$$

The fixed support ratios give $1 \leq \kappa_N \ll 1$. For a parameter R_0 in the admissible TPC-45 range, the inherited global prime band is

$$\mathcal{P}_{R_0}^\# := \left\{ p \in \mathbb{P} : \sqrt{N_+} < p \leq \frac{N_-}{R_0} \right\}. \quad (92)$$

Let y be very-high admissible. In particular, $y > \sqrt{N_+}$, and put

$$R_{\text{safe}}(y) := \frac{N_-}{2y}, \quad R_{\text{max}}(y) := \frac{N_+}{y} = 2\kappa_N R_{\text{safe}}(y). \quad (93)$$

These are the exact TPC-59 safe and maximal cofactor scales [8].

Define the portion of the native high carrier selected solely by the prime band condition:

$$\mathcal{W}_{R_0}^{\text{band}}(y) := \{w \in \mathcal{W}_H(y) : p(w) \in \mathcal{P}_{R_0}^\#\}. \quad (94)$$

All other row, orbit, opposite-row, mask, profile, and terminal labels are those already present in $\mathcal{W}_H(y)$; no favorable support is added.

Theorem 6.1 (Exact single-band/high-face sandwich). *For every very-high admissible y ,*

$$\boxed{\begin{aligned} &\{w \in \mathcal{W}_H(y) : \kappa_N R_0 \leq r(w) < R_{\text{max}}(y)\} \\ &\quad \subseteq \mathcal{W}_{R_0}^{\text{band}}(y) \\ &\quad \subseteq \{w \in \mathcal{W}_H(y) : R_0 \leq r(w) < R_{\text{max}}(y)\}. \end{aligned}} \quad (95)$$

Thus the only uncertainty created by the target-width constants lies in the fixed-ratio transition window

$$R_0 \leq r < \kappa_N R_0. \quad (96)$$

In particular, every high atom with $r < R_0$ is absent from the single band, while every high atom with $r \geq \kappa_N R_0$ is present in it.

Proof. If $w \in \mathcal{W}_{R_0}^{\text{band}}(y)$, then $p(w) \leq N_-/R_0$ and $n(w) \geq N_-$, so

$$r(w) = \frac{n(w)}{p(w)} \geq R_0. \quad (97)$$

Also $p(w) > y$ and $n(w) \leq N_+$, hence

$$r(w) < \frac{N_+}{y} = R_{\max}(y). \quad (98)$$

This proves the right inclusion.

Conversely, let $w \in \mathcal{W}_H(y)$ and suppose $r(w) \geq \kappa_N R_0$. Then

$$p(w) = \frac{n(w)}{r(w)} \leq \frac{N_+}{\kappa_N R_0} = \frac{N_-}{R_0}. \quad (99)$$

The very-high inequality $p(w) > y > \sqrt{N_+}$ supplies the lower endpoint of (92). Thus $p(w)$ lies in the band. The upper bound $r(w) < R_{\max}(y)$ is automatic on the high face. $\square$

Corollary 6.2 (Empty, top-window, and multiscale regimes). *The sandwich has the following exact consequences.*

- (i) If $R_0 \geq R_{\max}(y)$, then $\mathcal{W}_{R_0}^{\text{band}}(y) = \emptyset$.
- (ii) If $R_0 \asymp R_{\max}(y)$, the intersection is confined to $O(1)$ top dyadic cofactor blocks.
- (iii) If $R_0 = X^{\theta_0+o(1)}$ and $R_{\max}(y) = X^{\theta_J+o(1)}$ with $0 \leq \theta_0 < \theta_J$, then the band may contain high blocks of exponent $\theta \geq \theta_0$, but it omits every full block with $2R < R_0$.

These are support statements. In cases (ii) and (iii), the permitted blocks may still carry zero actual atomic mass.

6.2 The endpoint mismatch with the inherited freezing certificate

At the critical cutoff of TPC-59,

$$y = C_{\text{vh}}Q, \quad R_{\max}(y) \asymp J = X^{\theta_J+o(1)}, \quad \theta_J := \frac{133}{400}. \quad (100)$$

The balanced global band of TPC-45 has

$$R_0 = \ell_+ = X^{\lambda_L+o(1)}, \quad \lambda_L := \frac{99979}{210000}, \quad (101)$$

and therefore

$$\lambda_L - \theta_J = \frac{15077}{105000} > 0. \quad (102)$$

Corollary 6.3 (The balanced band misses the critical high face). *For all sufficiently large X , the balanced band satisfies*

$$R_0 > R_{\max}(C_{\text{vh}}Q), \quad \boxed{\mathcal{W}_{R_0}^{\text{band}}(C_{\text{vh}}Q) = \emptyset.} \quad (103)$$

This is not peculiar to the exact balanced choice. Write $R_0 = X^{\theta_0+o(1)}$. The sufficient TPC-45 band/cutoff freezing condition has the form

$$(\lambda_L - \theta_0)_+ + \lambda_0 \left(1 - \frac{1}{A}\right) < \frac{1}{400}, \quad A > 1, \quad (104)$$

where the second term is nonnegative [1]. Hence every band certified by this inequality obeys

$$\theta_0 > \lambda_L - \frac{1}{400} = \theta_J + \frac{29629}{210000}. \quad (105)$$

Theorem 6.4 (Single-band endpoint incompatibility). *At $y = C_{\text{vh}}Q$, every single band certified by (104) is eventually disjoint from the critical very-high face. Conversely, retuning $R_0 = X^{\theta_J+o(1)}$ to meet the top critical cofactor window incurs the coefficient-blind band exponent*

$$\lambda_L - \theta_J = \frac{15077}{105000} > \frac{1}{400}, \quad (106)$$

so the inherited sufficient freezing certificate no longer closes.

Proof. Equation (105) gives $R_0/R_{\max}(y) = X^{c+o(1)}$ for a fixed $c > 0$. Apply corollary 6.2(i). For the retuned value, substitution of $\theta_0 = \theta_J$ in the first term of (104) gives (106); adding the nonnegative cutoff term cannot improve the inequality. $\square$

The theorem closes one simultaneous use of the old support-only freezing certificate and the critical high face. It does not say that a different coefficient-specific freezing theorem is impossible.

6.3 Full omitted blocks and the harmonic zero-saving theorem

Suppose now that the band is retuned so that $R_0 < R_{\max}(y)$. The exact sandwich still forces every cofactor $r < R_0$ to be absent from the band-generated relation. Let $[R, 2R)$ be a full dyadic block satisfying

$$1 \leq R < 2R \leq R_0, \quad R \leq R_{\max}(y). \quad (107)$$

For a row m , call the block *fully omitted* when

$$[R, 2R) \cap \mathbb{N} \subseteq E_m := \{r \leq R_{\max}(y) : r \notin \mathcal{I}_m\}. \quad (108)$$

The TPC-61 harmonic capacity is

$$\mathfrak{h}_m(E_m) := \frac{m}{\varphi(m)} \sum_{r \in E_m} \left(1 + \frac{J/r}{\log(2 + J/r)}\right), \quad (109)$$

and its effective high exposure is

$$H_m^{\text{eff}} = \frac{W_H(m)}{U_m}, \quad 0 < H_m^{\text{eff}} \leq \#\mathcal{J}_X \ll J. \quad (110)$$

Theorem 6.5 (Harmonic zero-saving on a full omitted block). *Assume (108) on an active row. Then*

$$\boxed{\mathfrak{h}_m(E_m) \gg R + \frac{J}{\log(2 + J/R)}}. \quad (111)$$

Consequently,

$$\boxed{\frac{\mathfrak{h}_m(E_m)}{H_m^{\text{eff}}} \gg \frac{R}{J} + \frac{1}{\log(2 + J/R)}}. \quad (112)$$

If

$$J = X^{\theta_J+o(1)}, \quad R = X^{\theta+o(1)}, \quad 0 \leq \theta \leq \theta_J, \quad (113)$$

then the right side of (112) is $X^{-o(1)}$. Therefore the TPC-61 harmonic sufficient condition cannot certify

$$\frac{\mathfrak{h}_m(E_m)}{H_m^{\text{eff}}} \leq X^{-\sigma+o(1)} \quad (114)$$

for any fixed $\sigma > 0$.

Proof. The factor $m/\varphi(m)$ is at least one. There are $\gg R$ integers in $[R, 2R)$. Each contributes one to (109). Moreover, for $R \leq r < 2R$,

$$\frac{J/r}{\log(2 + J/r)} \geq \frac{J/(2R)}{\log(2 + J/R)}. \quad (115)$$

Summing over the block proves (111). Divide by $H_m^{\text{eff}} \ll J$ to obtain (112). Under (113), either $R/J = X^{-o(1)}$, or $\log(2 + J/R) = O(\log X)$; in both cases the displayed lower bound is $X^{-o(1)}$, which is larger than $X^{-\sigma+o(1)}$ for every fixed positive σ . $\square$

The same conclusion appears in the exponent law of TPC-61. A full block has cardinality exponent $e(\theta) = \theta$, while necessarily $H_m^{\text{eff}} \leq X^{\theta_J+o(1)}$. Even at the optimal balanced exposure exponent $\eta_H = \theta_J$, its certified saving is

$$\sigma_{\text{cof}}(\theta) = \eta_H - e(\theta) - (\theta_J - \theta) = 0. \quad (116)$$

Any smaller effective-exposure exponent only worsens the certificate.

Remark 6.6 (A certificate barrier, not an actual lower bound). TPC-61 proves the one-sided implication

$$\frac{W_M(m)}{W_H(m)} \ll \frac{\mathfrak{h}_m(E_m)}{H_m^{\text{eff}}}. \quad (117)$$

A lower bound for the right side does not lower-bound the left side. The omitted block may contain no realized largest-prime cofactors, or its literal weights may vanish. [Theorem 6.5](#) says only that the declared support-cardinality/Brun–Titchmarsh certificate cannot prove a fixed-power upper saving for a full omitted block. A weighted largest-prime-factor distribution theorem, an actual terminal-incidence estimate, or signed cancellation could still do so.

The window sandwich and endpoint exponent comparison are L0 arithmetic support statements specialized at L1 to the literal fixed- h_0 target interval and prime band. The harmonic lower calculation closes one L1 support-only certificate. None of these results proves a positive high-face mass, a lower terminal square, a Möbius correlation estimate, or an L2 fixed-shift parity improvement.

7 Labeled multiscale completion and the erasure gate

The single-band obstruction does not force one to sum different cofactor scales coherently. There is a canonical finite construction which first retains the scale as an orthogonal label. This construction is exact at the raw-energy level. It is important, however, not to identify it with the later physical output, where that label is absent.

7.1 The canonical dyadic direct sum

Let Ω_X be the coefficient-independent finite literal incidence which is being completed. Thus an atom $\omega \in \Omega_X$ carries, in particular, its recovered cofactor $r(\omega)$, and its pair projection is the declared canonical relation (in the maximal eligible application, this is $\mathcal{I}_X^{\text{max}}$). No coefficient-dependent zero is removed from Ω_X .

For $k \in \mathbb{Z}$, set

$$B_k = [2^k, 2^{k+1}) \cap \mathbb{N}, \quad \Omega_{X,k} := \{\omega \in \Omega_X : r(\omega) \in B_k\}, \quad (118)$$

and retain only the nonempty labels

$$K_X^{\text{dyad}} := \{k : \Omega_{X,k} \neq \emptyset\}, \quad L_X := |K_X^{\text{dyad}}|. \quad (119)$$

If $1 \leq r(\omega) \leq X^{C+o(1)}$ for one fixed C , as in the inherited polynomial cofactor ranges, then

$$L_X \leq 2 + \log_2 \left(\frac{\max_{\omega \in \Omega_X} r(\omega)}{\min_{\omega \in \Omega_X} r(\omega)} \right) = O(\log X) = X^{o(1)}. \quad (120)$$

Empty Ω_X is harmless and may be treated separately; below we assume it is nonempty.

Define

$$\mathcal{H}_X^{\text{raw}} := \ell^2(\Omega_X), \quad \mathcal{H}_{X,k} := \ell^2(\Omega_{X,k}), \quad \mathcal{H}_X^{\text{lab}} := \bigoplus_{k \in K_X^{\text{dyad}}} \mathcal{H}_{X,k}, \quad (121)$$

all with raw counting-measure atomic norms. The restriction map is

$$J_X : \mathcal{H}_X^{\text{raw}} \longrightarrow \mathcal{H}_X^{\text{lab}}, \quad J_X x = (x|_{\Omega_{X,k}})_{k \in K_X^{\text{dyad}}}. \quad (122)$$

Theorem 7.1 (Exact labeled multiscale completion). *The map J_X is unitary and, for every raw atom vector x ,*

$$\|J_X x\|_{\mathcal{H}_X^{\text{lab}}}^2 = \sum_{k \in K_X^{\text{dyad}}} \|x|_{\Omega_{X,k}}\|_2^2 = \|x\|_{\mathcal{H}_X^{\text{raw}}}^2. \quad (123)$$

At the pair level, writing $\mathcal{I}_{X,k}^{\text{max}} := \{(m, r) \in \mathcal{I}_X^{\text{max}} : r \in B_k\}$, one likewise has the disjoint identity

$$\mathcal{I}_X^{\text{max}} = \coprod_{k \in K_X^{\text{dyad}}} \mathcal{I}_{X,k}^{\text{max}}. \quad (124)$$

Consequently the passage to labeled dyadic blocks has exactly zero raw energy loss and zero fixed-power bookkeeping cost.

Proof. The sets in (118) are pairwise disjoint and their union is Ω_X . Parseval's identity for this coordinate partition gives (123); the inverse to J_X simply forgets the external block notation while retaining every atom. The pair identity follows from the same disjoint partition by the recovered cofactor. Finally, (120) implies $L_X^A = X^{o(1)}$ for every fixed A , while the exact Hilbert identity itself pays no factor at all. $\square$

The theorem neither creates an arithmetically active atom nor asserts that every block has a physical terminal realization. It says only that once a literal completed incidence has been declared, separating its cofactor scales loses no raw mass. In particular, the union in (124) must not be substituted for a coefficientwise physical identity that has been proved only on one of its blocks.

7.2 Exact analysis of label erasure

We now isolate the additional map hidden by the phrase “forget the labels.” Let Ξ_X be a finite output-coordinate set, let $\pi : \Omega_X \rightarrow \Xi_X$ be the declared grouping map, and attach a nonzero complex weight a_ω to every atom. Define

$$\mathcal{E}_a : \ell^2(\Omega_X) \longrightarrow \ell^2(\Xi_X), \quad (\mathcal{E}_a x)(\xi) := \sum_{\omega \in F_\xi} a_\omega x(\omega), \quad F_\xi := \pi^{-1}(\{\xi\}). \quad (125)$$

Coordinates with empty fiber carry no output information; henceforth they are deleted, so every F_ξ is nonempty. Put

$$s_\xi := \sum_{\omega \in F_\xi} |a_\omega|^2. \quad (126)$$

Theorem 7.2 (Fiber spectrum, norm, and kernel tests). *For the grouping map in (125),*

$$\mathcal{E}_a \mathcal{E}_a^* = \text{diag}(s_\xi)_{\xi \in \Xi_X}, \quad (127)$$

$$\|\mathcal{E}_a\|^2 = \max_{\xi: F_\xi \neq \emptyset} s_\xi, \quad (128)$$

$$\text{Ker } \mathcal{E}_a = \left\{ x : \sum_{\omega \in F_\xi} a_\omega x(\omega) = 0 \text{ for every } \xi \right\}, \quad (129)$$

$$\text{rank } \mathcal{E}_a = \#\{\xi : F_\xi \neq \emptyset\}, \quad \dim \text{Ker } \mathcal{E}_a = \sum_{\xi} (|F_\xi| - 1). \quad (130)$$

The nonzero squared singular values are exactly the numbers s_ξ . In particular, the full-domain lower constant

$$\eta_a := \inf_{x \neq 0} \frac{\|\mathcal{E}_a x\|_2^2}{\|x\|_2^2} \quad (131)$$

is positive if and only if every nonempty fiber is a singleton. In that case

$$\eta_a = \min_{\omega \in \Omega_X} |a_\omega|^2; \quad (132)$$

if some fiber contains at least two atoms, then $\eta_a = 0$.

Proof. The adjoint is $(\mathcal{E}_a^* y)(\omega) = \overline{a_\omega} y(\pi(\omega))$. Hence $(\mathcal{E}_a \mathcal{E}_a^* y)(\xi) = s_\xi y(\xi)$, proving (127) and (128). The displayed fiber equations give the kernel. A nonempty fiber contributes one to the rank and $|F_\xi| - 1$ to the nullity, proving (130). The positive spectrum of $\mathcal{E}_a^* \mathcal{E}_a$ equals that of $\mathcal{E}_a \mathcal{E}_a^*$. If a fiber contains two atoms, a nonzero weighted difference supported on those atoms lies in the kernel. If every fiber is a singleton, the Gram is diagonal with entries $|a_\omega|^2$, which proves (132). $\square$

For the literal unweighted erasure $a_\omega = 1$, the exact formulas specialize to

$$\|\mathcal{E}_1\|^2 = \max_{\xi} |F_\xi|, \quad \text{Ker } \mathcal{E}_1 = \left\{ x : \sum_{\omega \in F_\xi} x(\omega) = 0 \text{ for every } \xi \right\}. \quad (133)$$

If the only discarded coordinate is the dyadic block label and each block contributes at most one atom to a given output coordinate, then

$$\|\mathcal{E}_1\|^2 \leq L_X = O(\log X) = X^{o(1)}. \quad (134)$$

This is an upper norm, not a lower reassembly estimate. Two blocks which land on the same physical coordinate already give the kernel vector $(1, -1)$ on that fiber.

Normalizing each fiber by $|F_\xi|^{-1/2}$ would make the nonzero singular values equal to one, but it would neither remove the fiber kernel nor preserve the literal raw atomic normalization. Such a normalization therefore cannot be silently inserted into the physical packet.

7.3 The relevant test is on the actual upstream range

The full labeled space may be larger than the image of the actual coefficient map. Let $\mathbf{A}_X^{\text{lab}} : V_X^{\text{coef}} \rightarrow \mathcal{H}_X^{\text{lab}}$ be any declared upstream analysis map and put $S_X = \text{Ran } \mathbf{A}_X^{\text{lab}}$. Assume $S_X \neq \{0\}$, and define

$$\eta(\mathcal{E}_a; S_X) := \inf_{0 \neq v \in S_X} \frac{\|\mathcal{E}_a v\|_2^2}{\|v\|_2^2}. \quad (135)$$

Proposition 7.3 (Exact range and composition criteria). *In finite dimension,*

$$\eta(\mathcal{E}_a; S_X) > 0 \iff S_X \cap \text{Ker } \mathcal{E}_a = \{0\}, \quad (136)$$

and $\eta(\mathcal{E}_a; S_X)$ is the least eigenvalue of the compression of $\mathcal{E}_a^* \mathcal{E}_a$ to S_X . If

$$\|\mathbf{A}_X^{\text{lab}} z\|_2^2 \geq \alpha_X \|z\|_2^2, \quad \eta(\mathcal{E}_a; S_X) \geq \eta_X, \quad (137)$$

then

$$\|\mathcal{E}_a \mathbf{A}_X^{\text{lab}} z\|_2^2 \geq \eta_X \alpha_X \|z\|_2^2. \quad (138)$$

Conversely, a nonzero z with $\mathbf{A}_X^{\text{lab}} z \in \text{Ker } \mathcal{E}_a$ kills every full-coefficient lower bound for this particular composition.

Proof. The unit sphere of S_X is compact. The continuous function $v \mapsto \|\mathcal{E}_a v\|^2$ has positive minimum exactly when it has no zero there, which proves (136). Rayleigh–Ritz gives the compressed-Gram statement. Applying the two inequalities in (137) successively proves (138); the final assertion is immediate. $\square$

Thus the honest implication is

$$\boxed{\text{exact labeled raw completion} \not\Rightarrow \text{physical grouped lower bound.}} \quad (139)$$

One must first identify the literal grouping map and the actual upstream range, apply the kernel test, and only then estimate the compressed Gram. The logarithmic number of labels controls bookkeeping and certain upper norms; it does not control coherent cancellation after grouping.

8 Endpoint ledger and route decisions

The preceding results settle a provenance question and close one support-only certificate. They do not supply a new favorable exponent. This section records exactly how those facts enter the inherited endpoint calculus.

8.1 What the single-band calculation certifies

The critical row and cofactor scales remain

$$\theta_Q = \frac{267}{400}, \quad \theta_J = \frac{133}{400}, \quad \lambda_L = \frac{99979}{210000}. \quad (140)$$

The balanced inherited band has lower cofactor threshold $R_0 = X^{\lambda_L + o(1)}$, whereas the critical high face has ceiling $R_{\max} = X^{\theta_J + o(1)}$. Since

$$\lambda_L - \theta_J = \frac{15077}{105000} > \frac{1}{400}, \quad (141)$$

the balanced band, and in fact every band certified by the inherited freezing inequality, is eventually disjoint from the critical high face. Retuning R_0 down to the top critical scale makes the first inherited freezing loss equal to the left side of (141), already beyond the entire endpoint budget. Thus the old simultaneous band/freezing certificate is closed before exposure is estimated.

If a different theorem permits such a retuned band, every full omitted dyadic block below its cofactor window still faces a second obstruction: the available support-cardinality/Brun–Titchmarsh harmonic argument cancels its best possible cofactor saving. More precisely, if $\eta_H \leq \theta_J$ is its effective-exposure exponent, then

$$\sigma_{\text{cof}}^{\text{card}}(\eta_H) := \eta_H - \theta_J \leq 0, \quad \boxed{\sup_{\eta_H \leq \theta_J} (\sigma_{\text{cof}}^{\text{card}}(\eta_H))_+ = 0.} \quad (142)$$

Here $x_+ := \max\{x, 0\}$. Thus the certificate has no positive fixed-power saving. It cannot satisfy the strict sufficient condition

$$\sigma_M > \lambda_{\text{rep}} + \nu_M, \quad \lambda_{\text{rep}}, \nu_M \geq 0. \quad (143)$$

This kills the specified harmonic certificate on the full omitted blocks. Together with (141), it does not prove that the actual weighted missing mass has exponent zero, and it does not kill a weighted distribution estimate, a signed estimate, or the exact shorted-Gram route.

By contrast, the labeled completion in [theorem 7.1](#) has exact raw-energy constant one and only $O(\log X) = X^{o(1)}$ external labels. Its bookkeeping exponent is exactly zero. The erasure map is a different gate. If its actual range-restricted lower constant is

$$\eta(\mathcal{E}_a; S_X) \geq X^{-\lambda_{\text{erase}} + o(1)}, \quad (144)$$

then λ_{erase} is charged once inside the physical reassembly loss. Neither L_X nor λ_{erase} may be charged again under another name.

8.2 The strict complete ledger

Retaining the module names of TPC–61–TPC–62, every nonnegative fixed-power loss on the actual physical chain must occur once in

$$\Lambda_{\text{tot}} := \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{reasm}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}}. \quad (145)$$

The endpoint gate is

$$\Lambda_{\text{tot}} < \frac{1}{400}. \quad (146)$$

Here λ_{reasm} includes every genuine loss from labeled represented output to the declared physical output, including λ_{erase} when erasure is a mandatory factor. Exact provenance identities, set partitions, canonical relabelings, and the logarithmic block count cost zero. The quantities σ_M and ν_M determine whether a chosen mass–degree certificate makes reassembly perturbative; they are not extra summands in (145).

All estimates in the ledger must hold for the same fixed h_0 , the same literal carrier, the same raw atomic normalization, and the same uniform or distinguished coefficient scope. Equality at $1/400$ is rejected unless a separate leading-constant argument recovers a strict final inequality. No such constant-level recovery is proved here.

8.3 Decision tree with exact kill rules

The next attack should follow the order below.

- Gate 1.** *Pair faithfulness.* Apply the pre-terminal, terminal, and grouped rectangularization tests to the literal full incidence. If pair projection creates a terminal-active spurious cross, retain the full incidence hypergraph; do not force a pair-only model.
- Gate 2.** *Alignment first.* Compute the parent-to-terminal alignment mass $W_A(m)$. A row with $W_A(m) = 0 < W_H(m)$ kills a uniform activation theorem on every coefficient class containing that row. Enlarging the retained pair relation cannot repair this zero.
- Gate 3.** *Relation completion.* On aligned mass, compute the exact completion ratio. Passing to $\mathcal{I}_X^{\text{max}}$ removes only the relation-induced defect; it does not improve the alignment ratio.

- Gate 4.** *Labeled multiscale realization.* Retain every nonempty cofactor block as an orthogonal label. This passes the raw-energy gate exactly, but each newly included block still needs its own literal coefficient and terminal compatibility theorem.
- Gate 5.** *Erasur kernels.* Identify the actual upstream range S_X and compute $S_X \cap \text{Ker } \mathcal{E}_a$. A nonzero intersection kills the full-coefficient lower bound for that declared physical composition. A kernel in the ambient labeled space but outside S_X does not kill the composition.
- Gate 6.** *Weighted exposure or exact shorting.* The single-band inherited freezing certificate is incompatible with the critical face by (141). After any independently justified retuning, the cardinality proof is still closed by (142). Either prove a new weighted terminal/largest-prime-factor estimate, or use the exact signed or shorted-Gram route. Do not rename label count as missing mass saving.
- Gate 7.** *Represented and downstream chain.* Only after activation, represented native transfer, and reassembly are quantified in one scope should boundary and tail arrows be attached and (146) tested.

The exact deaths and surviving pivots can be summarized as follows.

Observed event	What is killed	Honest pivot
No pair-only terminal-faithful relation exists	Pair-rectangular completion of the literal incidence.	Retain the full incidence hypergraph.
The inherited freezing inequality certifies the single band	Its intersection with the critical high face: the certified band is eventually empty.	Prove a new band/freezing theorem or realize a genuinely multiscale carrier.
Retuning the old band to $R_0 \asymp J$	The inherited sufficient freezing route: its first loss already exceeds $1/400$.	A new coefficient-sensitive freezing estimate is required.
$W_A(m_0) = 0 < W_H(m_0)$ on a required row	Uniform parent-to-terminal activation on a class containing that row.	A justified distinguished class may avoid the row; otherwise reopen the terminal carrier.
The canonical completion ratio is one but W_A/W_H is small	The assertion that relation completion alone repairs exposure.	Prove terminal alignment or a weighted alignment theorem.
$S_X \cap \text{Ker } \mathcal{E}_a \neq \{0\}$	Uniform lower transfer through that actual erasure composition.	Change the physical representation, restrict to a justified distinguished vector, or estimate a different exact shorted map.
Best possible positive-part cardinality saving is zero	The support-cardinality/harmonic proof of the strict product test.	Weighted distribution, signed cancellation, or exact shorting.
A necessary proved loss reaches $1/400$	The stated endpoint exponent route without constant-level recovery.	Change the architecture; do not absorb the loss into $X^{o(1)}$.

8.4 Forbidden substitutions

None of these gates is repaired by specializing h_0 to 2, averaging over shifts, deleting a coefficient-dependent zero, normalizing raw atoms by fiber size, choosing phases after seeing the coefficient vector, or replacing the literal active support by its terminal image. Those operations change

either the object or the quantifier. Their conclusions cannot be entered into (145) for the present fixed carrier.

9 Result ledger, claim boundary, and the next gate

TPC-63 completes a missing support-provenance interface and proves that one inherited single-band certificate cannot deliver a fixed-power exposure saving. It does not prove that the underlying arithmetic mass is bad. The distinction is the main conclusion of this paper.

9.1 Result ledger

Object	Exact conclusion	Status
Literal provenance	The upstream retained support is the coefficient-independent projection of a full incidence; preservation of already supported pairs does not prove exposure on an independently declared native-high carrier.	L0 logic; L1 audit
Rectangular completion	Pre-terminal raw-form, terminal-atomic, and grouped-operator faithfulness have different finite incidence tests; spurious crosses are explicit.	L0 iff tests
Canonical pair relation	When pair-only terminal completion is possible there is a unique least terminal-faithful relation; otherwise the full incidence hypergraph is the faithful replacement.	L0 theorem
Maximal eligible core	Every declared retained relation selects the same native-high carrier as its intersection with the coefficient-independent maximal eligible relation; rectangular terminal output still needs the faithfulness test.	L0 identity
Exposure factorization	Row exposure is exactly the product of parent-to-terminal alignment and retained completion inside the aligned measure; the corresponding uniform and distinguished constants are sharp minimax quantities.	L0 theorem; L1 specialization
Single-band geometry	The exact sandwich is $R_0 \leq r < R_{\max}(y)$, up to one fixed-ratio transition window. The balanced band, and every band certified by the inherited freezing inequality, is eventually disjoint from the critical high face; retuning to $R_0 \asymp J$ makes the inherited freezing loss exceed $1/400$.	L1 support no-go
Support-only barrier	After any independently justified retuning, every fully omitted lower dyadic block leaves the inherited support-cardinality/Brun–Titchmarsh harmonic certificate with no positive fixed-power saving, so it cannot pass the strict mass-degree test.	L1 certificate no-go
Labeled completion	Orthogonal direct sum over all nonempty dyadic cofactor blocks preserves raw energy exactly and uses $O(\log X) = X^{o(1)}$ labels.	L0 theorem
Erasure/grouping	The fiber square masses give the exact operator norm and nonzero singular values; fiber equations give the exact kernel and range-restricted injectivity test.	L0 theorem

Object	Exact conclusion	Status
Endpoint calculus	Provenance and logarithmic labeling cost zero; every genuine erasure or physical loss is charged once, and the complete total must be strictly below $1/400$.	L0 ledger; L1 use conditional

9.2 The L0/L1/L2 boundary

Level	Established here	Not licensed by that level
L0	Finite incidence identities, canonical/minimal support constructions, exact exposure minimax laws, orthogonal block energy, and grouping spectrum/kernel tests.	No assertion about prime distribution, signed Möbius cancellation, or physical fixed-shift reassembly.
L1	Specialization to one literal fixed nonzero h_0 , its actual coefficient-free support, raw atomic norm, cofactor window, and the failure of the named support-only certificate.	No positive fixed-power missing-mass estimate, represented lower bound, boundary/tail closure, or number-theoretic target lower bound.
L2	None.	In particular, no signed fixed- h_0 parity-kernel estimate, fixed $h_0 = 2$ prime-pair estimate, prime-pair asymptotic, or twin-prime implication.

The L1 negative result concerns a proof mechanism, not the truth of the desired weighted estimate. Calling it an “exposure barrier” means that support cardinality and the inherited harmonic upper bound are insufficient on the omitted blocks. It does not mean those blocks carry a large proportion of the actual raw weight.

9.3 What remains unproved

The following inputs remain open.

1. The parent-to-terminal alignment ratio is not shown to be positive, much less $1 - X^{-\sigma+o(1)}$, on every active physical row.
2. The single-band low-cofactor complement is not shown to have a fixed-power weighted saving in either uniform or distinguished scope.
3. The additional dyadic blocks in the labeled completion are not shown automatically to inherit the literal coefficient identity, terminal activation, or physical output formula of the original band.
4. The actual physical grouping map and its upstream range are not shown to pass the kernel test or to have an endpoint-quality least singular value.
5. There is no single carrier on which critical occupancy, singleton survival, represented transfer, alias control, boundary reconnection, and the tail estimate are all proved.
6. No term in the strict endpoint ledger is newly assigned a favorable positive margin by this paper.

In particular, setting $\mathcal{I}_X = \mathcal{I}_X^{\max}$ closes only the relation completion factor. It cannot retroactively turn a terminal-inactive parent atom into an active one, and it cannot identify a labeled direct sum with an unlabeled physical sum.

9.4 The next theorem: terminal activation in a fixed scope

The next useful paper should attack the alignment factor before spending a difficult cancellation estimate on a possibly empty carrier. There are two quantitatively different targets.

Positive activation target. For a full-coefficient statement, prove with one fixed $\lambda_{\text{act}} \geq 0$

$$\inf_{m: W_H(m) > 0} \frac{W_A(m)}{W_H(m)} \geq X^{-\lambda_{\text{act}} + o(1)}. \quad (147)$$

For a declared distinguished coefficient vector z^* , the honest alternative is

$$\frac{\sum_m W_A(m) |z_m^*|^2}{\sum_m W_H(m) |z_m^*|^2} \geq X^{-\lambda_{\text{act},*} + o(1)}. \quad (148)$$

The second statement must not be promoted to the first.

Near-full alignment target. The perturbative missing-mass route needs the stronger estimate

$$\sum_m (W_H(m) - W_A(m)) |z_m|^2 \leq X^{-\sigma_{\text{align}} + o(1)} \sum_m W_H(m) |z_m|^2, \quad \sigma_{\text{align}} > 0, \quad (149)$$

in the same uniform or distinguished scope used downstream. A merely positive proportion does not imply (149).

There are then two honest construction routes.

- Route 1.** *Preserve the single band.* Prove a genuinely weighted terminal or largest-prime-factor estimate on its low-cofactor complement. Reusing only support cardinality and the old harmonic bound cannot work by (142).
- Route 2.** *Realize the multiscale completion.* Construct the literal coefficient and terminal maps on every added dyadic block, retain the block label, and then compute the exact physical erasure on the actual upstream range. The set union and Parseval identity do not supply these arrows.

Either route should begin with an exact zero test. If a required active row has $W_A = 0 < W_H$, the uniform activation route is closed and that no-go is itself the correct result. If activation survives, the following layer should establish a represented-reference Gram lower bound on the same carrier before returning to determinant collisions and signed alias cancellation.

Conclusion. The provenance gap has now been converted into two separate, testable questions: does the literal terminal predicate activate enough of the native parent mass, and can the completed labeled carrier be mapped back to the physical output without an active-range kernel? TPC-63 answers neither by definition. It proves the exact finite calculus, shows why the old single-band cardinality proof cannot answer the first question, and specifies the next activation theorem without crossing the L1/L2 claim boundary.

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